

Math 2250 (Elementary Linear Algebra) Section 03

Syllabus—Spring 2019

Instructor: Bryce Christopherson
Office: Ross Hall 303
Email: bchris19@uwyo.edu

Office Hours: RH 303, W, 3:10 PM - 4:00 PM
MAC Hours: RH 029, F, 12:00 PM - 2:00 PM
Class: MWF 2:10 PM – 3:00 PM in AS 210

Prerequisites: A grade of a C or better in MATH 2200 or MATH 2350.

Textbook: *Linear Algebra and its Applications*, fifth edition, by D.C. Lay, S.R. Lay, and J.J. MacDonald.

Course Description: In this class, we will cover the following topics: linear equations and matrices, vector spaces, linear transformations, determinants, orthogonality, eigenvalues and eigenvectors.

Exams: All exams will be administered outside the regularly scheduled class time (see below). Attendance is required; a make-up exam will only be administered if there is documentation from a proper authority, such as a note from a physician in the case of illness. If you know that you will miss an exam, please talk to your instructor at least one week before the date the exam will be administered. If you have any questions about the scheduling of the exams, please discuss this with your instructor.

On Exams, you will be permitted to use a scientific non-graphing calculator. Graphing calculators and notes may not be used on exams under any circumstances. Midterm Exams I and II, as well as the Final Exam, are scheduled at the dates/times listed below. Locations will be announced later during the semester.

Homework: Eleven written homework assignments will be given during the course of the semester. There is **one deadline** associated to each assignment: the **check date**.

- (i) Before the beginning of class on the **check date**, you must submit a either a typed or handwritten and scanned copy of your first attempt at the assignment through the class WyoCourses page. Your instructor will verify that you have made what can reasonably be considered to be a good-faith effort to complete the assignment. If you have done so, your instructor will score your assignment accordingly to verify this. If a first draft is not submitted by the check date, you will not receive credit for the assignment.
- (ii) After class on the **check date** of each assignment, a solution key to the assignment will be posted on the class WyoCourses page. Using the posted solutions, you may check and correct your work as you see fit.

To receive full credit on your submissions, you are **required** to show **all** of your work in a **neat, coherent**, and **mathematically correct** fashion. **Late assignments will not be accepted under any circumstances without a university approved exception.**

Grading Policy: Your percentage grade is determined by the following:

Exam I	220 pts (20%)	Thursday, 03/07/2019, 5:15 PM - 7:00 PM
Exam II	220 pts (20%)	Thursday, 04/11/2019, 5:15 PM - 7:00 PM
Final Exam	440 pts (40%)	Friday, 05/17/2019, 1:15 PM - 3:15 PM
Written Homework	220 pts (20%)	
Total	1100 pts (100%)	

You can estimate your letter grade by using the following scale: 90–100% is an A, 80–89% is a B, 70–79% is a C, 60–69% is a D, and 0–59% is an F. Plus/Minus Grades will not be awarded, except in very rare circumstances upon the discretion of the instructor.

Extra Credit: Opportunities for earning extra credit may occur throughout the semester at the discretion of the instructor. Extra credit problems may be given in class or occasionally may be offered on exams. Points earned from extra credit may help borderline grades, but do not expect extra credit to significantly change your final score.

Attendance: Attendance to regularly scheduled lecture sessions is not mandatory, but I believe it is very important. If you frequently do not attend lecture without an exceptionally good reason and begin to struggle in the course, I will have very little sympathy for you.

Getting Help: One of the most important skills a student can develop is the ability to recognize quickly when they are struggling with some concept in a class. Once you recognize you need help, first check through your materials, such as class notes, to see if the answer is there. If you cannot alleviate whatever difficulties you are having yourself, *please seek help*. There are lots of resources available to you to help you succeed.

Study Groups: Meet with some of your classmates and help each other with questions. Learning from your peers is one of the best ways to learn.

Office Hours: My office hours (subject to change as noted above) are Wednesday, 3:10 PM - 4:00 PM in RH 303 and Friday, 12:00 PM - 2:00 PM at the Math Assistance Center (MAC) in RH 029. If you find yourself struggling with the course, please try to take advantage of office hours. If these times are not convenient, we should be able to arrange an alternative time. I am here to help you.

The Math Assistance Center(MAC): The MAC is a free drop-in tutoring center for students enrolled in 1000 and 2000 level math courses. It is located in Ross Hall 29 (northwest corner, on bottom floor). They are open Monday - Thursday 10:00 AM - 5:00 PM and Friday 10:00 AM - 1:00 PM. It's also a great place to study, work on homework, or meet with study groups.

STEP: Tutoring is available in this class at the STEP Tutor Center. STEP is open Sunday-Thursday from 5:00 PM - 9:00 PM and is located in Coe Library. Sessions are drop in (no appointments) and are 30 minutes in length. STEP has also opened a Satellite Tutor Center in the basement of Washakie and has online tutoring available as well. Please visit the STEP Tutor Schedule for days and times: www.uwyo.edu/step. STEP opens Sunday, February 3rd.

Walk-In Academic Coaching The Center for Advising and Career Services in 222 Knight Hall offers walk-in coaching to any student. They can help on a variety of topics, such as Note Taking, Time Management, Exam Preparation, Textbook Reading, Staying Motivated, Managing Finals, etc.

It is important that you try a variety of different services. If you do not receive the help you need at that time, please try it again or try a different service. Don't be shy – make sure you get the help you need!

Academic Dishonesty and Classroom Conduct: The University of Wyoming is built upon a strong foundation of integrity, respect and trust. All members of the university community have a responsibility to be honest and the right to expect honesty from others. Any form of academic dishonesty (see UW Regulation 6-802) is unacceptable to our community and will not be tolerated.

You are expected to avoid any behaviors that would be disruptive in class. I reserve the right to ask you to leave or to put away any devices that are not helpful should I deem it necessary. Persistence in such behavior may get you dropped from the course. Refer to the A&S College policy document *Students and Teachers—Working Together*.

Disability Statement: The University of Wyoming is committed to providing equitable access to learning opportunities for all students. If you have a disability, including but not limited to physical, learning, sensory or psychological disabilities, and would like to request accommodations in this course due to your disability, please register with and provide documentation of your disability as soon as possible to Disability Support Services (DSS), Room 128 Knight Hall. You may also contact DSS at (307) 766-3073 or udss@uwyo.edu. It is in the students best interest to request accommodations within the first week of classes, understanding that accommodations are not retroactive. Visit the DSS website for more information at: www.uwyo.edu/udss

Additionally, if you feel that you have what people refer to as *math anxiety*, I urge you to call or visit the Counseling Center in room 341, Knight Hall and get some help. Their phone number is (307) 766-2187.

Changes to Syllabus: The policies listed in this syllabus are subject to change. Minor changes will be announced in class and substantive changes shall be communicated in writing.

Grading Rubric: The rubric below will be used *as a guideline* when evaluating your written work on homework assignments and exams throughout this course. You should strive to produce written solutions that are clear and well organized, make proper use of mathematical notation and terminology, demonstrate an understanding of the mathematical concepts involved, and result in a correct solution. If you practice these skills while working on your written homework problems, you will be more likely to meet the standard expected of your written work on exams. Please familiarize yourself with this rubric and reference it if ever you feel your work has been unfairly graded.

Characteristics of the Written Work	Qualitative Description	Typical Scoring %
(i) The work is presented in a neat, clear, organized fashion that is complete and easy to follow. (ii) Graphs are correct, complete, carefully drawn and appropriately labeled. (iii) Correct terminology and notation are always used. (iv) The solution shows a complete understanding of the mathematical concepts involved, although it may contain a minor arithmetic error or sign mistake.	Exceptional	90-100
(i) The work may not be completely clear or well organized, or it may have some steps missing, but it is still possible to follow with little effort. (ii) Graphs are correct and complete, but somewhat carelessly drawn or missing some labels. (iii) Correct terminology and notation are usually used. (iv) The solution shows almost complete understanding of the mathematical concepts and may contain one or two arithmetic or sign errors.	Good	75-90
(i) The work may be unclear, poorly organized, or have several steps missing, but it is possible to deduce the intended method of solution with some effort. (ii) Graphs are partially correct, somewhat carelessly drawn, and may be missing some labels. (iii) Correct terminology and notation are sometimes used. (iv) The solution shows a limited understanding of the mathematical concepts involved, or an almost complete understanding of the mathematical concepts involved, but contains several arithmetic or sign errors.	Fair	60-75
(i) The work is poorly organized or is very incomplete, making it difficult to follow. (ii) Graphs are incorrect, carelessly drawn, or incomplete, and may not be labeled. (iii) Correct terminology and notation are seldom used. (iv) The solution shows a limited understanding of the mathematical concepts involved, and may also contain several arithmetic or sign errors.	Poor	40-60
(i) The work is so disorganized or incomplete that it is very difficult or impossible to follow. (ii) Graphs are not drawn or are completely wrong and not labeled. (iii) There is little or no correct usage of terminology and notation. (iv) The solution shows little or no understanding of the mathematical concepts involved.	Very Poor	0-40

Course Schedule: (Sections Covered: 1.1-1.5, 1.7-1.9, 2.1-2.5, 2.8, 2.9, 3.1-3.3, 4.1-4.7, 5.1-5.5, 5.7, 6.1-6.5)

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
27 Jan.	28 Jan.	29 Jan.	30 Jan.	31 Jan.	1 Feb.	2 Feb.
	Intro		1.1		1.2	
3 Feb.	4 Feb.	5 Feb.	6 Feb.	7 Feb.	8 Feb.	9 Feb.
	1.3		1.4		1.5	
					HW #1 (Check)	
10 Feb.	11 Feb.	12 Feb.	13 Feb.	14 Feb.	15 Feb.	16 Feb.
	1.7		1.8		1.9	
					HW #2 (Check)	
17 Feb.	18 Feb.	19 Feb.	20 Feb.	21 Feb.	22 Feb.	23 Feb.
	2.1		2.2		2.3	
					HW #3 (Check)	
24 Feb.	25 Feb.	26 Feb.	27 Feb.	28 Feb.	1 Mar.	2 Mar.
	2.4		2.5		2.8	
					HW #4 (Check)	
3 Mar.	4 Mar.	5 Mar.	6 Mar.	7 Mar.	8 Mar.	9 Mar.
	2.9		Review	Exam I (1.1-1.5, 1.7-1.9, 2.1-2.5)	3.1	
10 Mar.	11 Mar.	12 Mar.	13 Mar.	14 Mar.	15 Mar.	16 Mar.
	3.2		3.3		3.3	
					HW #5 (Check)	
17 Mar.	18 Mar.	19 Mar.	20 Mar.	21 Mar.	22 Mar.	23 Mar.
	Spring Break	Spring Break	Spring Break	Spring Break	Spring Break	

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
24 Mar.	25 Mar.	26 Mar.	27 Mar.	28 Mar.	29 Mar.	30 Mar.
	4.1		4.2		4.3	
31 Mar.	1 Apr.	2 Apr.	3 Apr.	4 Apr.	5 Apr.	6 Apr.
	4.4		4.5		4.6 HW #6 (Check)	
7 Apr.	8 Apr.	9 Apr.	10 Apr.	11 Apr.	12 Apr.	13 Apr.
	4.7		Review	Exam II (2.8-2.9,3.1-3.3, 4.1-4.7)	5.1 HW #7 (Check)	
14 Apr.	15 Apr.	16 Apr.	17 Apr.	18 Apr.	19 Apr.	20 Apr.
	5.2		5.3		5.4 HW #8 (Check)	
21 Apr.	22 Apr.	23 Apr.	24 Apr.	25 Apr.	26 Apr.	27 Apr.
	5.5		5.7		6.1 HW #9 (Check)	
28 Apr.	29 Apr.	30 Apr.	1 May.	2 May.	3 May.	4 May.
	6.2		6.3		6.4 HW #10 (Check)	
5 May.	6 May.	7 May.	8 May.	9 May.	10 May.	11 May.
	6.5		Review HW #11 (Check)		Last Day of Class	
12 May.	13 May.	14 May.	15 May.	16 May.	17 May.	18 May.
					Final Exam (5.1-5.5,5.7, 6.1-6.5)	